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Introduction to Data Science

Lecture 9 Statistics

Maximum Likelihood Estimate

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Probability

- **Knowing** the true **model**, you can **predict** the **samples** outcome.

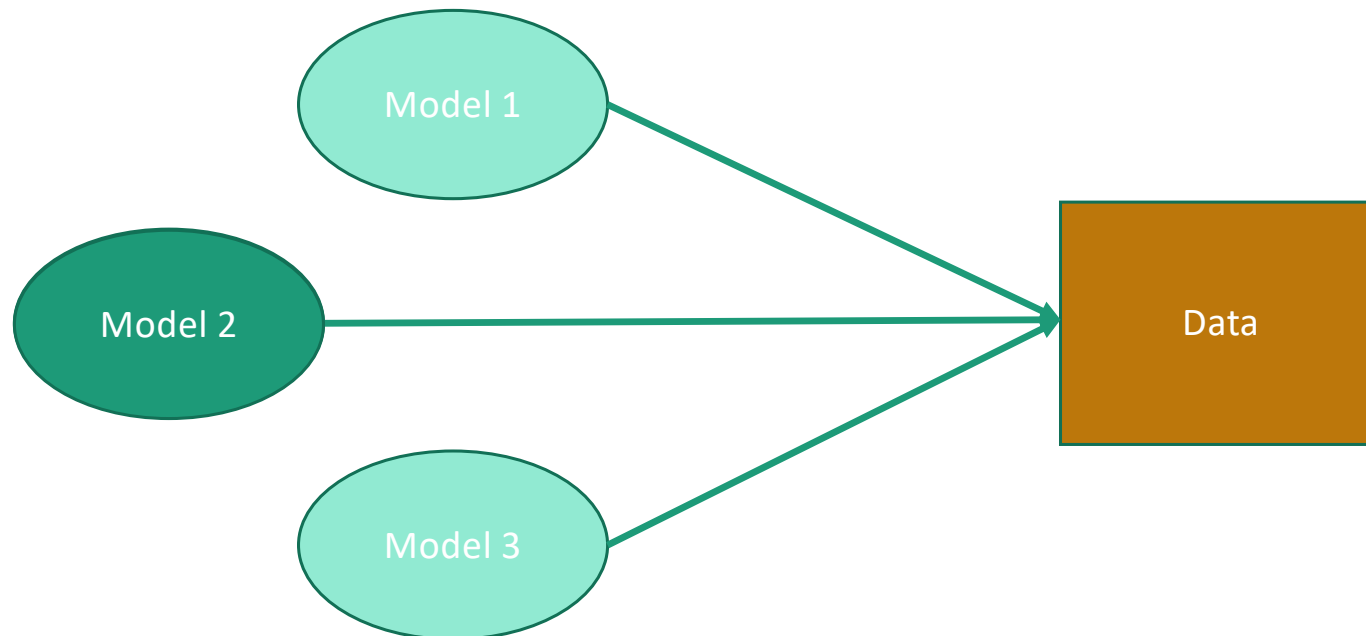
Statistics

- **Knowing** the generated **samples**, you can **estimate** the true **model**.

The general setup

Example: We have a data set, and we want to **infer** the properties of the underlying distribution from this data set.

- A variety of different probability models must at least be explored
- To see which model best “fits” the data.



In other words, you need

- Samples
- A set of possible models
- Criterion for quantifying model performance

Reading Materials

- Applied Statistics and Probability for Engineers, Third Edition, Douglas C. Montgomery and George C. Runger.
- Chap 7-3.2: Method of Maximum Likelihood

Example 1

- Whether a drug can cure a disease is a Bernoulli RV
- Yes ($X=1$) w.p. p ; no ($X = 0$) w.p. $1 - p$.
- p is unknown.
- 100 Samples: 81 of them were clinically successful.
- What's your estimation of p ? 0.81

Example 1

- Whether a drug can cure a disease is a Bernoulli RV
- Yes ($X=1$) w.p. p ; no ($X = 0$) w.p. $1 - p$.
- p is unknown.
- N Samples: M of them were clinically successful.
- What's your estimation of p ? M/N

Definition

- Given samples: $X_1, X_2, \dots, X_n$
- Goal: estimate an unknown parameter θ of the true model, (we use samples).

The hat notation

The point estimator used to estimate a parameter θ is usually denoted as $\hat{\theta}$.

Definition

- Given samples: $X_1, X_2, \dots, X_n$
- Goal: estimate an unknown parameter θ of the true model, (we use samples).
- The estimate $\hat{\theta}$ clearly depends on $(X_1, X_2, \dots, X_n)$, why?
 - $\hat{\theta}$ is a function of samples $(X_1, X_2, \dots, X_n)$
 - $\hat{\theta}$ extracts some (useful) information from the samples.
- For example, you can use $\bar{X} := \frac{\sum_i X_i}{N}$ (**sample mean**) as $\hat{\theta}$

Definition

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- Goal: estimate an unknown parameter θ of the true model, (we use samples).
- The estimate $\hat{\theta}$ clearly depends on $(X_1, X_2, \dots, X_n)$, why?
 - $\hat{\theta}$ is a function of samples $(X_1, X_2, \dots, X_n)$
 - $\hat{\theta}$ extracts some (useful) information from the samples.
- And we call $\hat{\theta}$ the **statistic**.

How to choose statistic?

- Whether a drug can cure a disease is a Bernoulli RV
- Yes ($X=1$) w.p. p ; no ($X = 0$) w.p. $1 - p$.
- N Samples: M of them were clinically successful.
- What's your estimation of p ? M/N

Why you use $\frac{M}{N}$ to estimate p ?

Example 2

- Drug 1: $p = 0.8$
- Drug 2: $p = 0.9$
- Samples: $M = 81$ successes from $N=100$ experiments.
- Which drug do you think is experimented?

How to determine the best parameter?

- Among **those possible choices**, choose the one that is **most likely** to generate the samples we have.

Example 1

- For drug experiments,
 - **N** experiments
 - **M** successes
- Possible models: Bernoulli distribution with probability p .
- Given one model, the probability of generating such samples:
 $L(p) = p^M (1 - p)^{N-M}$
- Most likely: the model **maximizes** the probability $L(p)$

Example 1

- Given one model, the probability of generating such samples:

$$L(p) = p^M (1 - p)^{N-M}$$

- Maximize the probability



We often maximize the **logarithm** of the probability (usually easier to maximize) of generating such samples

$$l(p) = \log L(p) = M * \log p + (N-M) * \log (1-p)$$

Example 1

- To maximize a continuous function $l(p)$
 - Find the point $l'(p) = 0$ and $l''(p) < 0$
 - Before we discuss optimization, all exercises with continuous cases just require you to show the point $l'(p) = 0$.

$$l(p) = \log L(p) = M * \log p + (N-M) * \log (1-p)$$

$$l'(p) = \frac{M}{p} - \frac{N-M}{1-p} = 0 \quad \longrightarrow \quad p = M/N$$

Formal Definition

- Given a model with an unknown parameter θ
- Given samples: $X_1, X_2 \dots, X_n$
- The probability that the models generates the samples is called **likelihood**. $L(\theta) = P(X_1, X_2 \dots, X_n | \theta)$
- To determine the best θ , we choose $\hat{\theta}$ such that $L(\theta)$ is maximized at $\theta = \hat{\theta}$. **Maximum likelihood estimate (MLE)**

Formal Definition

- Define $l(\theta) = \log L(\theta)$, which will be called log-**likelihood**.
 - In our course, log represents natural logarithm
- MLE: choose $\hat{\theta}$ such that $l(\theta)$ is maximized at $\theta = \hat{\theta}$

Likelihood function

- Given a model with an unknown parameter θ
- Given samples: $X_1, X_2 \dots, X_n$

Continuous RV model:

- Likelihood: $L(\theta) = \prod_i f(X_i | \theta)$
- Log-Likelihood: $l(\theta) = \sum_i \log(f(X_i | \theta))$

f : the model's probability density function (PDF)

Discrete RV model:

- Likelihood: $L(\theta) = \prod_i P(X_i | \theta)$
- Log-Likelihood: $l(\theta) = \sum_i \log(P(X_i | \theta))$

P : the model's probability mass function (PMF)

Likelihood function

- Given a model with an unknown parameter θ
- Given samples: $X_1, X_2 \dots, X_n$

Continuous RV model: $l(\theta) = \sum_i \log(f(X_i | \theta))$

Discrete RV model: $l(\theta) = \sum_i \log(P(X_i | \theta))$

Possible models: $\theta \in \Theta$ (additional information)

- Whether Θ are discrete or continuous.

Example 3: $\Theta = [0, \infty)$

- Suppose lifetime of each bulb is an exponential distribution.
 - Given parameter λ , pdf is $f(x) = \lambda e^{-\lambda x}$
- Samples of lifetime: $X_1, X_2, \dots, X_n$
- $L(\lambda) = \lambda^n e^{-\lambda \sum_i X_i}$
- $l(\lambda) = \text{Log } L(\lambda) = n \log \lambda - \lambda \sum_i X_i$
- $l'(\lambda) = \frac{n}{\lambda} - \sum_i X_i = 0 \quad \longrightarrow \quad \hat{\lambda} = n / \sum_i X_i = 1/\bar{X}.$

Example 4: $\Theta = (-\infty, \infty)$

- Suppose heights of people follow a normal distribution.
 - Given parameter μ , the model is $N(\mu, 1)$ Mean μ , variance 1
- Samples: $X_1, X_2, \dots, X_n$
- $L(\mu) = \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}\sum_i (X_i - \mu)^2}$
- $l(\mu) = \text{Log } L(\mu) = -\frac{1}{2}\sum_i (X_i - \mu)^2 - \frac{n}{2}\log 2\pi$
- $l'(\mu) = \sum_i (X_i - \mu) = 0 \quad \longrightarrow \quad \hat{\mu} = \bar{X}.$

Example 5: discrete Θ

- You may have additional information.
 - Drug 1: $p = 0.8$
 - Drug 2: $p = 0.9$
 - Samples: $M=81$ successes from $N = 100$ experiments.
- Potential models: Bernoulli with $p \in \{0.8, 0.9\}$

$$l = \log L = M \cdot \log p + (N-M) \cdot \log (1-p)$$

- Drug 1: -48.6539
- Drug 2: -52.2833

Conclusion: drug 1 is more likely to be experimented.